

A guide to manufacturing execution system (MES) | by ERP information | Medium

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A guide to manufacturing execution system (MES) To enhance your business, you need appropriate resource planning, a good quality management system, productive utilization of tools on the shop floor, effective HR management, etc. The manufacturing execution system is structured to fulfill all the needs. Read out the article to get more information about the manufacturing execution system (MES). A manufacturing execution system is a factory floor information and communication system with several functional capabilities. This system is essential for successful manufacturing. It helps to control, monitor, and track the entire manufacturing process. Simultaneously it helps to control inventory, equipment, and employees. It lies between the enterprise level and tool level control system.

Functionalities of MES It includes functions such as - Resource allocation and status- Operation/detailed scheduling- Dispatching production units- Document control- Data collection and acquisition- Labor management- Quality management- Process management- Maintenance management- Product tracking and genealogy- Performance analysis. It can provide feedback from the factory floor on a real-time basis. It interfaces with and complements ERP systems.

Advantages of Manufacturing Execution System - This process is useful for all type of industries including semiconductor, automotive, pharma companies, aerospace etc- Gathers useful data on the production line. Hence you can upgrade your manufacturing process- Allows you to manage your machines and employees in an easier way- It helps to eliminate the human error in the manufacturing process- It monitors the manufacturing activities throughout the industry and link them in lead time- Helps to control the manufacturing supply chain- Improves the quality of the product by providing real-time quality data checks- Eradicates valueless activities like paperwork- Helps to find the faults easily- Supports complicated workflow process- Reduces manufacturing cycle time and data entry time- Helps to improve customer service

History of Manufacturing Execution System Around the 1970s some of the manufacturing industries started using software applications to automate their accounts. Time files and these applications improved and started providing particular features for standard inventory management. In the early 1980s, manufacturing requirements planning (MRP) came with material planning, material control, and material monitoring. In the 1990s, the system improved and the manufacturing execution system comes into the picture.

Best practices for MES implementation - Define clear rules to implement the project.- To define the rules properly involve the members of the production, quality, execution, maintenance, and IT department throughout the rules-making process.- Involve stakeholders during the implementation process.- Set an achievable schedule for implementation and testing.- Originate a detailed MES system as per the user requirements with a distance between the control layer and the enterprise resource planning (ERP) layer.- Perform risk management for design, development, and deployment stages.- Determine the merging points between the layers to comprise your architecture. For example, MES stands for execution and ERP stands for planning. So ERP directs the MES what to make and MES tells what was made and where it went.- Combine the MES software with other production control systems.

Common mistakes during implementation - Not giving attention to the connection of MES requirements with particular business drivers.- Skipping the steps while getting into the details of master data ownership, interface point to the control system in the designing stage.- Improper determination of user requirements.- Not reviewing the design before implementation. If you review the design before only, you can make the changes before the implementation.- Improper establishment of the existing operating system. If you do not establish the current operating system, it is difficult to compare and find out the benefits of the MES system.- Not evaluating the system after implementation. Before switching over completely

to the new system, It is necessary to test the system even you implement it as per the plan. Good software for Manufacturing Execution System - Odoo- Mar-kov- Quantum- Tulip platform- Tavant Warranty- MIE Trak pro- Pro Shop- Genius ERP- Prodsmart Difference between ERP and MES The major difference is ERP being used to create and manage your basic schedules for production, material use, delivery, shipping and helps to collect information for your business. ERP system is a mixture of many modules like production, supply chain, order processing, inventory management, customer relationship management, and human resource management. Usually, it concentrates on working within the time limit of months, weeks, and days. But MES helps to monitor and track your shop floor manufacturing process and also allows you to generate reports on production plant activities in real-time. Get more definitions about manufacturing execution system (MES) and other ERP-related terms [here](#).